Development effects of electrification in Brazil

Identification method and replication of main results of *Lipscomb et al (2013)*

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Research design: ideal benchmark vs. the paper

- **Ideal design**: a lottery decides which feasible areas get the grid now and which wait ten years (cluster-randomized for spillovers).
- **The paper**: county and decade FE-IV (2SLS), 1960-2000, where the instrument is a *simulated grid* built only from geography and each decade's budget and assigned based on budget.

Causal interpretation only if

Relevance: met, first-stage $F \approx 25$
Independence: no time-varying confounder follows the grid (Table 10)
Exclusion: geography affects development only via electricity
Monotonicity: plausible, not discussed

Main findings

Preferred IV specification (county and decade FE, Amazon $\times$ decade dummies):

- **Housing values**: +10 pp electrification $\rightarrow$ **+881 reais**. It's a change of +6.8% of the mean value (Table 6).
- **HDI**: moving from 0 to full electrification $\rightarrow$ **+0.11**. Would move the median county towards the 69th percentile county (Table 7).

Mechanisms: electrification appears to improve development through education and employment gains, with limited evidence that migration explains the results and no effect on health.

Replication of main results

Is the design convincing?

Convincing

The instrument simulates electricity expansion using hydropower geography and decade-specific budgets, leaving out local demand. Results remain robust after additional controls for the geographic factors used to construct the instrument (Table 10).

Not convincing

The instrument predicts dams: they bring construction, roads, flood control and ecosystem change. This could bring positive (or negative!) effects on economic development, unrelated to electrification. Could this risk the exclusion restriction?